

Managers, Entrepreneurs, and the Allocation of Talent: Evidence from Hungary's Transition

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June 24, 2026

Acknowledgements

Acknowledgements



**Funded by
the European Union**



European Research Council
Established by the European Commission



**NEMZETI
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This research was funded by the European Research Council (ERC Advanced Grant agreement number 101097789) and by the National Research, Development and Innovation Office (Forefront Research Excellence Program contract number 144193). The views expressed in this project are those of the authors and do not necessary reflect the official view of the European Union, the European Research Council, or the National Research, Development and Innovation Office.

Introduction

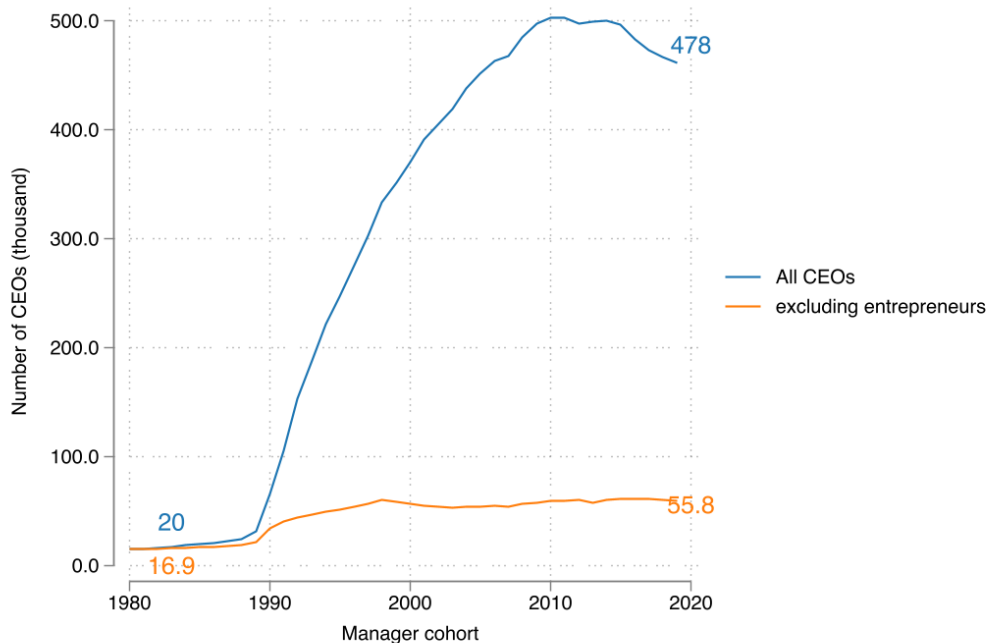
Hungary, 1980 (Fortepan / Szalay Zoltán)



Hungary, 1990 (MTI)



Number of Executive Positions Increased



What Can We Learn From Hungary?

Use Hungarian post-socialist transition as a natural experiment to study the supply side of the market for managers.

Why Micro $\neq$ Macro

What we know

- 1 Management matters
- 2 Training works
- 3 Managers matter

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What we need

- 1 Endogenous supply: how to incentivize people to become managers?
- 2 Selection: who will become managers?
- 3 Competition: what are the GE feedbacks of interventions?

Setup and Data

Data

Manager Data 1985-2019

Universe of corporations (1m) and their CEOs (1.3m).

Firm Financials 1985-2019

Firm size (employment) as proxy for manager quality.

Outline

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- 1 A steady-state model of manager selection
- 2 Managers and entrepreneurs
- 3 Calibration and counterfactuals

An Equilibrium Model of Managers

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- 1 Managers differ in innate skill
- 2 Returns to manager skill subject to corporate governance friction
- 3 Self-selection into management
- 4 Wages determined in equilibrium

Production Function

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Aggregate GDP:

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with sum of manager skills

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Policy goal: Increase Z via either N (more managers) or $\bar{z}$ (better training).

Corporate Governance Friction

Operating surplus,

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Owners cannot commit to sharing more than a fraction of surplus.

Manager wage is

$$\omega(z) \leq \phi \Pi(z) = \phi z \pi(w)$$

with $\phi < 1$.

Underprovision of manager skills.

Career Choice and Skill Supply

Manager if $\omega(z) > w$,

$$z > z_{\min}(Z).$$

Supply of manager skills is Pareto

$$\Pr(z > z_{\min}) = z_{\min}^{-\theta} z_0^{\theta}$$

$$E(z) := z_* = \frac{\theta}{\theta - 1} z_0$$

Labor Market Clearing

$$N + L_p = L$$

$$n := \frac{N}{L}$$

Aggregate GDP

$$\frac{Y}{L} = (Az_*)^\nu n^{\nu(1-1/\theta)} (1-n)^{1-\nu}$$

Increasing n

- 1 adds to stock of manager skills
- 2 dampened by selection (average manager is worse)
- 3 reduces production employment

Equilibrium and First Best

First best

$$\frac{n^*}{1 - n^*} = \frac{\nu(1 - 1/\theta)}{1 - \nu}$$

Equilibrium

$$\frac{n}{1 - n} = \phi \frac{\nu(1 - 1/\theta)}{1 - \nu} < \frac{n^*}{1 - n^*}$$

Too few managers.

What Can Policy Do?

- 1 Reduce corporate governance frictions.
- 2 Subsidize managers.
- 3 Provide management training.

Managers and Entrepreneurs

Managers and Entrepreneurs

Suppose an α share of population has “entrepreneurial spirit”:

- 1 can start own firm
- 2 capture full rent from running it

Others work for other entrepreneurs, either as manager or as worker.

Aggregate GDP

$$\frac{Y}{L} = \left(\frac{\theta}{\theta - 1} \right)^{\nu} (Az_0)^{\nu} \left[\alpha n_1^{1-1/\theta} + (1 - \alpha) n_0^{1-1/\theta} \right]^{\nu} (1 - n)^{1-\nu}$$

Additional misallocation

$$n_0 < n^* < n_1$$

Too few managers, too many entrepreneurs. Uniform pro-manager policy may be harmful.

Taking the Model to the Data

Goal

Calibrate model to micro data (1992-2019).

Conduct policy counterfactuals.

Table 1: Model Parameter Calibration

Parameter	Description	Target Moment	Preferred Value
ϕ	Manager rent share	Entrepreneur firm size	0.60
θ	Selection parameter	Cohort selection elasticity	11.0
A_M/A_E	Relative productivity	Within-person entrepreneur discount	1.78
ν	Management elasticity	CEO share in workforce	0.16
α	Entrepreneurial spirit	Share of entrepreneurs among CEOs	0.86

Managers and Entrepreneurs

Table 2: Manager Entry by Cohort

Cohort	Non-Entrepreneur	Entrepreneur	Total
1990	58,725	155,988	214,713
1995	59,058	156,117	215,175
2000	64,990	148,526	213,516
2005	66,804	118,981	185,785
2010	74,485	106,447	180,932
2015	56,202	54,840	111,042
2020	36,573	44,647	81,220
Total	427,234	785,546	1,212,780

Notes: This table reports the number of managers entering each cohort by entrepreneurship status. Cohorts are defined by the first year a manager appears in the data, grouped in 5-year bins. Entrepreneurs are defined as CEOs who are also founders of the firm they manage. The sample is restricted to Hungarian managers with non-missing revenue data and at most 4 simultaneous positions.

Entrepreneurs are less productive

Table 3: Entrepreneur Discount

	Dependent Variable: Log Revenue		
	(1) No Controls	(2) With Controls	(3) CEO FE
Entrepreneur CEO	-0.460*** (0.005)	-0.430*** (0.005)	-0.577*** (0.008)
CEO Age		0.112*** (0.002)	0.176*** (0.003)
CEO Age Squared		-0.001*** (0.000)	-0.002*** (0.000)
Firm Age		0.082*** (0.001)	0.047*** (0.001)
Firm Age Squared		-0.003*** (0.000)	-0.002*** (0.000)
Constant	9.749***	6.861***	5.066***

Large cohorts are less productive

Table 4: Manager Selection and Cohort Entry Effects

	Dependent Variable: Log Revenue				
	(1) Baseline	(2) No Founders	(3) Post-1992	(4) Post-2004	(5) Firm
$(\ln n)$					
Number of CEO Entrants, log	-0.137*** (0.019)	-0.104*** (0.019)	-0.091*** (0.015)	-0.055*** (0.017)	-0.03*** (0.017)
CEO Age	0.126*** (0.012)	0.162*** (0.011)	0.112*** (0.010)	0.124*** (0.010)	0.08*** (0.010)
CEO Age Squared	-0.001*** (0.000)	-0.002*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Entrepreneur CEO	-0.415*** (0.037)	0.000 (.)	-0.412*** (0.036)	-0.362*** (0.034)	-0.13*** (0.034)
Firm Age	0.081***	0.024***	0.076***	0.055***	0.07***

Policy Counterfactuals

The manager-only economy

Table 5: Counterfactual Policies in the M-Economy

	Baseline	First best	10% sub- sidy	Training
Manager subsidy	—	0.667	0.100	—
GDP per capita	1.000	1.015	1.005	1.016
Managers	0.097	0.151	0.105	0.097
Avg. skill	1.000	0.960	0.992	1.100
Worker wage	1.000	1.080	1.014	1.016
Manager wage	1.000	1.125	1.022	1.016

The M-E economy

Table 6: Counterfactual Policies in the E-M Economy

	Baseline	First best	10% sub- sidy	Second best
Manager subsidy	—	0.667	0.100	0.168
Entrepreneur subsidy	—	—	—	0.168
GDP per capita	1.000	1.040	1.016	1.001
Entrepreneurs	0.097	0.002	0.064	0.114
Managers	0.032	0.150	0.061	0.038
Avg. E skill	1.000	1.454	1.041	0.988
Avg. M skill	1.000	0.872	0.946	0.988
Worker wage	1.000	1.063	1.007	1.024
E wage	1.000	0.731	0.967	1.037
M wage	1.000	1.219	1.064	1.037

Conclusion

Conclusion

- Tractable, quantifiable model of manager demand and supply.
- Novel data for Hungary, 1985-2019.
- Use transition as macro shock to identify macro model.

Appendix

Literature

- **Large-scale management interventions:** Italy (Giorcelli 2019); US WW II (Bianchi & Giorcelli 2022); US business-school expansion (Giorcelli 2023)
- **Large-scale education interventions:** Italy (Bianchi & Giorcelli 2020), Colombia (Ferreyra et al 2023), Vietnam (Vu 2023)
- **Selection of talent:** Denmark (Akcigit, Pearce & Prato 2020)
- **Calibrated models with skill supply + selection:** Guner et al 2008; Bhattacharya et al 2013; Gomes & Kuehn 2017; Esfahani 2019; Akcigit, Alp & Peters 2021; Engbom et al 2024